

3-2 Assignment: Identifying CIFAR-10 Images

To Begin: Run all code blocks and observe the output. Once you have reviewed the sample output. Use the **LastName_FirstName_Assignment2.ipynb** file to complete your assignment.

Note: For compatability purposes, libraries have been updated from those used in the required readings to match to current versions; hence some of the package invocations may differ slightly from the book. The affected lines of code have comments added to the right as applicable, with the old code commented out above for reference.

GPU/CUDA/Memory Warnings/Errors: You may receive some errors referencing that GPUs will not be used, CUDA could not be found, or free system memory allocation errors. These and a few others, are standard errors that can be ignored here as they are environment based.

Example messages:

- Could not find cuda drivers on your machine, GPU will not be used.
- Please check linkage and avoid linking the same target more than once.
- E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
- Allocation of ##### exceeds 10% of free system memory

Installing Required Packages

This is to install necessary components to run the assignment

```
In [1]: !pip install -r requirements.txt
```

Requirement already satisfied: numpy==1.26 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from -r requirements.txt (line 1)) (1.26.0)

Requirement already satisfied: keras==3.5.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from -r requirements.txt (line 2)) (3.5.0)

Requirement already satisfied: Tensorflow==2.17 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from -r requirements.txt (line 3)) (2.17.0)

Requirement already satisfied: absl-py in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (2.1.0)

Requirement already satisfied: rich in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (13.7.1)

Requirement already satisfied: namex in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (0.0.8)

Requirement already satisfied: h5py in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (3.11.0)

Requirement already satisfied: optree in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (0.12.1)

Requirement already satisfied: ml-dtypes in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (0.4.0)

Requirement already satisfied: packaging in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from keras==3.5.0->-r requirements.txt (line 2)) (24.1)

Requirement already satisfied: astunparse>=1.6.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (1.6.3)

Requirement already satisfied: flatbuffers>=24.3.25 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (24.3.25)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (0.6.0)

Requirement already satisfied: google-pasta>=0.1.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (0.2.0)

Requirement already satisfied: libclang>=13.0.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (18.1.1)

Requirement already satisfied: opt-einsum>=2.3.2 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (3.3.0)

Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<5.0.0dev,>=3.20.3 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (4.25.3)

Requirement already satisfied: requests<3,>=2.21.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (2.32.3)

Requirement already satisfied: setuptools in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (71.1.0)

Requirement already satisfied: six>=1.12.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (1.16.0)

Requirement already satisfied: termcolor>=1.1.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (2.4.0)

Requirement already satisfied: typing-extensions>=3.6.6 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (4.12.2)

Requirement already satisfied: wrapt>=1.11.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (1.16.0)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (1.65.1)

Requirement already satisfied: tensorboard<2.18,>=2.17 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (2.17.0)

Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from Tensorflow==2.17->-r requirements.txt (line 3)) (0.37.1)

Requirement already satisfied: wheel<1.0,>=0.23.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from astunparse>=1.6.0->Tensorflow==2.17->-r requirements.txt (line 3)) (0.43.0)

Requirement already satisfied: charset-normalizer<4,>=2 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from requests<3,>=2.21.0->Tensorflow==2.17->-r requirements.txt (line 3)) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from requests<3,>=2.21.0->Tensorflow==2.17->-r requirements.txt (line 3)) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from requests<3,>=2.21.0->Tensorflow==2.17->-r requirements.txt (line 3)) (2.2.2)

Requirement already satisfied: certifi>=2017.4.17 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from requests<3,>=2.21.0->Tensorflow==2.17->-r requirements.txt (line 3)) (2024.7.4)

Requirement already satisfied: markdown>=2.6.8 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from tensorboard<2.18,>=2.17->Tensorflow==2.17->-r requirements.txt (line 3)) (3.6)

Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from tensorboard<2.18,>=2.17->Tensorflow==2.17->-r requirements.txt (line 3)) (0.7.2)

Requirement already satisfied: werkzeug>=1.0.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from tensorboard<2.18,>=2.17->Tensorflow==2.17->-r requirements.txt (line 3)) (3.0.3)

Requirement already satisfied: markdown-it-py>=2.2.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from rich->keras==3.5.0->-r requirements.txt (line 2)) (3.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from rich->keras==3.5.0->-r requirements.txt (line 2)) (2.18.0)

Requirement already satisfied: mdurl~0.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from markdown-it-py>=2.2.0->rich->keras==3.5.0->-r requirements.txt (line 2)) (0.1.2)

Requirement already satisfied: MarkupSafe>=2.1.1 in /home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages (from werkzeug>=1.0.1->tensorboard<2.18,>=2.17->Tensorflow==2.17->-r requirements.txt (line 3)) (2.1.5)

[notice] A new release of pip is available: 24.1.2 -> 26.1.1

[notice] To update, run: `pip install --upgrade pip`

```
In [2]: from keras.datasets import cifar10
        from keras.utils import to_categorical # Syntax change due to version bump
        from keras.models import Sequential
        from keras.layers import Dense, Dropout, Activation, Flatten # Syntax change due to
```

```

from keras.layers import Conv2D, MaxPooling2D
from keras.optimizers import SGD, Adam, RMSprop
import matplotlib.pyplot as plt
import tensorflow as tf
import os # For saving model purposes

```

Note: # Logs that appear below are informational!

```

In [3]: # CIFAR_10 is a set of 60K images 32 x 32 pixels on 3 channels
IMG_CHANNELS = 3
IMG_ROWS = 32
IMG_COLS = 32
os.environ["CUDA_VISIBLE_DEVICES"] = "-1"

# Constant
BATCH_SIZE = 100
NB_EPOCH = 20
NB_CLASSES = 10
VERBOSE = 1
VALIDATION_SPLIT = 0.2
OPTIM = RMSprop()

```

Note:

If an UNKNOWN ERROR LOG DISPLAYS, it's just tensorflow trying to connect to GPU.
It'll simply redirect and use the CPU!

```

In [4]: # Load the datasets
(X_train, y_train), (X_test, y_test) = cifar10.load_data()
print('X_train shape:', X_train.shape)
print(X_train.shape[0], 'train samples')
print(X_test.shape[0], 'test samples')

```

```

X_train shape: (50000, 32, 32, 3)
50000 train samples
10000 test samples

```

Note:

One-hot Encoding & Normalization of images

In the book, the older version is:

```
Y_train = np_utils.to_categorical(y_train, NB_CLASSES)
```

```

In [5]: Y_train = to_categorical(y_train, NB_CLASSES)
Y_test = to_categorical(y_test, NB_CLASSES)

# float and normalization
X_train = X_train.astype('float32')
X_test = X_test.astype('float32')
X_train /= 255
X_test /= 255

```

```
In [6]: # Network
model = Sequential()
model.add(Conv2D(32, (3, 3), padding="same",
                input_shape=(IMG_ROWS, IMG_COLS, IMG_CHANNELS)))
model.add(Activation('relu'))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Dropout(0.25))
model.add(Flatten())
model.add(Dense(512))
model.add(Activation('relu'))
model.add(Dropout(0.5))
model.add(Dense(NB_CLASSES))
model.add(Activation('softmax'))
model.summary()
# Below red is simply a userwarning.
```

/home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Par
conv2d (Conv2D)	(None, 32, 32, 32)	
activation (Activation)	(None, 32, 32, 32)	
max_pooling2d (MaxPooling2D)	(None, 16, 16, 32)	
dropout (Dropout)	(None, 16, 16, 32)	
flatten (Flatten)	(None, 8192)	
dense (Dense)	(None, 512)	4,194
activation_1 (Activation)	(None, 512)	
dropout_1 (Dropout)	(None, 512)	
dense_1 (Dense)	(None, 10)	5
activation_2 (Activation)	(None, 10)	

Total params: 4,200,842 (16.02 MB)

Trainable params: 4,200,842 (16.02 MB)

Non-trainable params: 0 (0.00 B)

Note:

Training...

NOTE: THIS MAY TAKE SOME TIME. Go grab a cup of coffee!

```
In [7]: model.compile(loss="categorical_crossentropy", optimizer=OPTIM, metrics=["accuracy"]
model.fit(X_train, Y_train, batch_size=BATCH_SIZE, epochs=NB_EPOCH, validation_spli
score = model.evaluate(X_test, Y_test, batch_size=BATCH_SIZE, verbose=VERBOSE)
print('Test Score:', score[0])
print('Test Accuracy:', score[1])
```

Epoch 1/20
400/400 ————— **12s** 28ms/step - accuracy: 0.3027 - loss: 1.9887 - val_accuracy: 0.4687 - val_loss: 1.4510

Epoch 2/20
400/400 ————— **11s** 27ms/step - accuracy: 0.4956 - loss: 1.4183 - val_accuracy: 0.5572 - val_loss: 1.2466

Epoch 3/20
400/400 ————— **11s** 27ms/step - accuracy: 0.5445 - loss: 1.2765 - val_accuracy: 0.5662 - val_loss: 1.2343

Epoch 4/20
400/400 ————— **11s** 27ms/step - accuracy: 0.5820 - loss: 1.1812 - val_accuracy: 0.6023 - val_loss: 1.1282

Epoch 5/20
400/400 ————— **11s** 27ms/step - accuracy: 0.6116 - loss: 1.1019 - val_accuracy: 0.6065 - val_loss: 1.1076

Epoch 6/20
400/400 ————— **11s** 27ms/step - accuracy: 0.6395 - loss: 1.0327 - val_accuracy: 0.6194 - val_loss: 1.1082

Epoch 7/20
400/400 ————— **11s** 26ms/step - accuracy: 0.6559 - loss: 0.9793 - val_accuracy: 0.6475 - val_loss: 1.0223

Epoch 8/20
400/400 ————— **11s** 26ms/step - accuracy: 0.6776 - loss: 0.9162 - val_accuracy: 0.6461 - val_loss: 1.0237

Epoch 9/20
400/400 ————— **11s** 26ms/step - accuracy: 0.6935 - loss: 0.8748 - val_accuracy: 0.6550 - val_loss: 0.9957

Epoch 10/20
400/400 ————— **10s** 26ms/step - accuracy: 0.7108 - loss: 0.8330 - val_accuracy: 0.6592 - val_loss: 1.0088

Epoch 11/20
400/400 ————— **11s** 26ms/step - accuracy: 0.7254 - loss: 0.7913 - val_accuracy: 0.6596 - val_loss: 1.0029

Epoch 12/20
400/400 ————— **10s** 26ms/step - accuracy: 0.7387 - loss: 0.7509 - val_accuracy: 0.6655 - val_loss: 1.0094

Epoch 13/20
400/400 ————— **11s** 26ms/step - accuracy: 0.7535 - loss: 0.7034 - val_accuracy: 0.6757 - val_loss: 0.9656

Epoch 14/20
400/400 ————— **11s** 26ms/step - accuracy: 0.7612 - loss: 0.6812 - val_accuracy: 0.6642 - val_loss: 1.0040

Epoch 15/20
400/400 ————— **11s** 27ms/step - accuracy: 0.7765 - loss: 0.6435 - val_accuracy: 0.6810 - val_loss: 0.9946

Epoch 16/20
400/400 ————— **11s** 26ms/step - accuracy: 0.7931 - loss: 0.6015 - val_accuracy: 0.6853 - val_loss: 0.9812

Epoch 17/20
400/400 ————— **11s** 26ms/step - accuracy: 0.7965 - loss: 0.5875 - val_accuracy: 0.6813 - val_loss: 0.9908

Epoch 18/20
400/400 ————— **11s** 26ms/step - accuracy: 0.8004 - loss: 0.5681 - val_accuracy: 0.6846 - val_loss: 1.0216

Epoch 19/20
400/400 ————— **11s** 26ms/step - accuracy: 0.8157 - loss: 0.5312 - val_a

```
ccuracy: 0.6822 - val_loss: 1.0160
Epoch 20/20
400/400 ----- 11s 27ms/step - accuracy: 0.8256 - loss: 0.5013 - val_a
ccuracy: 0.6873 - val_loss: 1.0011
100/100 ----- 1s 8ms/step - accuracy: 0.6766 - loss: 0.9925
Test Score: 1.0188692808151245
Test Accuracy: 0.6704000234603882
```

```
In [8]: # Save the model
os.makedirs('output', exist_ok=True)
model_json = model.to_json()
open('./output/cifar10_architecture.json', 'w').write(model_json)

# And the weights learned by our deep network on the training set
model.save_weights('./output/cifar10_weights.h5', overwrite=True) # NOTE, it is now
```

```
In [9]: # Improving the CIFAR-10 performance with deeper network
model = Sequential()
model.add(Conv2D(32, (3,3), padding='same', input_shape=(IMG_ROWS, IMG_COLS, IMG_CH
model.add(Activation('relu'))
model.add(Conv2D(32, (3, 3), padding='same'))
model.add(Activation('relu'))
model.add(MaxPooling2D(pool_size=(2,2)))
model.add(Dropout(0.25))
model.add(Conv2D(64, (3, 3), padding='same'))
model.add(Activation('relu'))
model.add(Conv2D(64, 3, 3))
model.add(Activation('relu'))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Dropout(0.25))
model.add(Flatten())
model.add(Dense(512))
model.add(Activation('relu'))
model.add(Dropout(0.5))
model.add(Dense(NB_CLASSES))
model.add(Activation('softmax'))

model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Par
conv2d_1 (Conv2D)	(None, 32, 32, 32)	
activation_3 (Activation)	(None, 32, 32, 32)	
conv2d_2 (Conv2D)	(None, 32, 32, 32)	9
activation_4 (Activation)	(None, 32, 32, 32)	
max_pooling2d_1 (MaxPooling2D)	(None, 16, 16, 32)	
dropout_2 (Dropout)	(None, 16, 16, 32)	
conv2d_3 (Conv2D)	(None, 16, 16, 64)	18
activation_5 (Activation)	(None, 16, 16, 64)	
conv2d_4 (Conv2D)	(None, 5, 5, 64)	36
activation_6 (Activation)	(None, 5, 5, 64)	
max_pooling2d_2 (MaxPooling2D)	(None, 2, 2, 64)	
dropout_3 (Dropout)	(None, 2, 2, 64)	
flatten_1 (Flatten)	(None, 256)	
dense_2 (Dense)	(None, 512)	131
activation_7 (Activation)	(None, 512)	
dropout_4 (Dropout)	(None, 512)	
dense_3 (Dense)	(None, 10)	5
activation_8 (Activation)	(None, 10)	

Total params: 202,282 (790.16 KB)

Trainable params: 202,282 (790.16 KB)

Non-trainable params: 0 (0.00 B)

```
In [10]: # Improving the CIFAR-10 performance with data augmentation
from tensorflow.keras.preprocessing.image import ImageDataGenerator # Note, we are
from keras.datasets import cifar10
import numpy as np
NUM_TO_AUGMENT = 5
```

```
In [11]: # Load dataset
(X_train, y_train), (X_test, y_test) = cifar10.load_data()

# Augmenting
print("Augmenting training set images...")
datagen = ImageDataGenerator(rotation_range=40,
```

```
width_shift_range=0.2,
height_shift_range=0.2,
zoom_range=0.2,
horizontal_flip=True,
fill_mode='nearest')
```

Augmenting training set images...

The Below will run to create images from the above ImageDataGenerator().

Note: If you run it, it will take awhile!

```
In [12]: xtas, ytas = [], []

for i in range(X_train.shape[0]):
    num_aug = 0
    x = X_train[i] #(3, 32, 32)
    x = x.reshape((1,) + x.shape) # (1 , 3, 32, 32)
    for x_aug in datagen.flow(x, batch_size=1, save_to_dir='preview', save_prefix='
        if num_aug >= NUM_TO_AUGMENT:
            break
        xtas.append(x_aug[0])
        num_aug += 1
```

```
In [13]: optimizer = RMSprop() # Recreating the optimizer

#for the datagen
datagen.fit(X_train)

model.compile(loss="categorical_crossentropy", optimizer=optimizer, metrics=["accu

# train changed from model.fit_generator()
history = model.fit(
    datagen.flow(X_train, Y_train, batch_size=BATCH_SIZE),
    steps_per_epoch=X_train.shape[0] // BATCH_SIZE,
    epochs=NB_EPOCH,
    validation_data=tf.data.Dataset.from_tensor_slices((X_test, Y_test)).batch(BATC
    verbose=VERBOSE
)

# score = model.evaluate(X_test, Y_test, batch_size=BATCH_SIZE, verbose=VERBOSE)

print("Test Score:", score[0])
print("Test Accuracy:", score[1])
```

Epoch 1/20

/home/codio/.pyenv/versions/3.11.9/lib/python3.11/site-packages/keras/src/trainers/d
ata_adapters/py_dataset_adapter.py:122: UserWarning: Your `PyDataset` class should c
all `super().\_\_init\_\_(\*\*kwargs)` in its constructor. `\*\*kwargs` can include `workers
, `use\_multiprocessing`, `max\_queue\_size`. Do not pass these arguments to `fit()`,
as they will be ignored.

self._warn_if_super_not_called()

500/500 ————— 39s 75ms/step - accuracy: 0.1334 - loss: 3.0978 - val_a
ccuracy: 0.2874 - val_loss: 1.9454

Epoch 2/20

```
/home/codio/.pyenv/versions/3.11.9/lib/python3.11/contextlib.py:158: UserWarning: Your input ran out of data; interrupting training. Make sure that your dataset or generator can generate at least `steps_per_epoch * epochs` batches. You may need to use the `.repeat()` function when building your dataset.  
    self.gen.throw(typ, value, traceback)
```

500/500  **1s** 3ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.2874 - val_loss: 1.9454
 Epoch 3/20
500/500  **37s** 74ms/step - accuracy: 0.2710 - loss: 1.9655 - val_accuracy: 0.3350 - val_loss: 1.8259
 Epoch 4/20
500/500  **1s** 3ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.3350 - val_loss: 1.8259
 Epoch 5/20
500/500  **37s** 74ms/step - accuracy: 0.3222 - loss: 1.8465 - val_accuracy: 0.3968 - val_loss: 1.6393
 Epoch 6/20
500/500  **1s** 3ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.3968 - val_loss: 1.6393
 Epoch 7/20
500/500  **38s** 76ms/step - accuracy: 0.3499 - loss: 1.7792 - val_accuracy: 0.4108 - val_loss: 1.6264
 Epoch 8/20
500/500  **1s** 3ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.4108 - val_loss: 1.6264
 Epoch 9/20
500/500  **38s** 76ms/step - accuracy: 0.3672 - loss: 1.7363 - val_accuracy: 0.3689 - val_loss: 1.8273
 Epoch 10/20
500/500  **1s** 3ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.3689 - val_loss: 1.8273
 Epoch 11/20
500/500  **37s** 73ms/step - accuracy: 0.3862 - loss: 1.6923 - val_accuracy: 0.4020 - val_loss: 1.6256
 Epoch 12/20
500/500  **1s** 2ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.4020 - val_loss: 1.6256
 Epoch 13/20
500/500  **36s** 70ms/step - accuracy: 0.3989 - loss: 1.6658 - val_accuracy: 0.4412 - val_loss: 1.5181
 Epoch 14/20
500/500  **1s** 2ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.4412 - val_loss: 1.5181
 Epoch 15/20
500/500  **35s** 69ms/step - accuracy: 0.4048 - loss: 1.6499 - val_accuracy: 0.4297 - val_loss: 1.5397
 Epoch 16/20
500/500  **1s** 2ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.4297 - val_loss: 1.5397
 Epoch 17/20
500/500  **35s** 69ms/step - accuracy: 0.4128 - loss: 1.6316 - val_accuracy: 0.4781 - val_loss: 1.4539
 Epoch 18/20
500/500  **1s** 2ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.4781 - val_loss: 1.4539
 Epoch 19/20
500/500  **35s** 70ms/step - accuracy: 0.4166 - loss: 1.6267 - val_accuracy: 0.4557 - val_loss: 1.4926
 Epoch 20/20
500/500  **1s** 2ms/step - accuracy: 0.0000e+00 - loss: 0.0000e+00 - val_accuracy: 0.4557 - val_loss: 1.4926

Test Score: 1.0188692808151245

Test Accuracy: 0.6704000234603882

MARKDOWN WORK HERE

- Analyze the ethical and privacy implications of the algorithm you just created here!

There are numerous ethical and privacy implications of the algorithm ran today. Because of the classification of animals based on such low quality images into broad categories it can easily lead to animals being misclassified. Furthermore an algorithm such as this can be used to kill or exterminate animals based on their classification driving extinction. Finally because it's a machine learning algorithm it requires a massive amount of data raising concerns about privacy and consent to being in data models such as this one.

In []: